

TechMart Data Analysis Report

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Introduction

TechMart is a growing retail chain operating across North America and Europe, with data spanning employee performance, product inventory, customer demographics, and sales transactions. This report analyzes those four interconnected datasets to answer three questions relevant to TechMart leadership: which store locations perform best and why, which products drive the most volume within their categories, and how customers behave once they start spending. The findings below are intended to support decisions around staffing, inventory planning, and loyalty program strategy.

Key Findings

Insight 1: Employee sales performance varies sharply by store location, and headcount does not predict performance.

Chicago posted the highest average sales performance at \$3,875, roughly 64% higher than the lowest-performing location, Houston, at \$2,363.64, despite Chicago having the smallest team in the dataset (8 employees vs. Houston's 11). Chicago also outperformed the seven-location average of \$2,983.04 by nearly 30%. This suggests either a concentration of high performers in Chicago or an efficiency gap worth investigating in Houston.

Location	Avg Sales	Employees	Rank
Chicago	\$3,875.00	8	1
Phoenix	\$3,312.50	16	2
London	\$3,214.29	14	3
New York	\$2,823.53	17	4
Los Angeles	\$2,692.31	13	5
Paris	\$2,600.00	10	6
Houston	\$2,363.64	11	7

Insight 2: Keyboards and tablets lead their categories, with a clearer drop-off in Accessories than in Electronics.

In Accessories, Keyboards led with 156 units sold, ahead of Chargers (123) and Speakers (85). In Electronics, Tablets led with 134 units, ahead of Mice (129) and Headphones (110). The Electronics top three are far more tightly clustered (134, 129, 110) than the Accessories top three (156, 123, 85), suggesting more balanced demand across top Electronics products versus a single dominant product in Accessories.

Category	Product	Total Sold	Rank in Category
Accessories	Keyboard	156	1
Accessories	Charger	123	2
Accessories	Speaker	85	3
Electronics	Tablet	134	1
Electronics	Mouse	129	2
Electronics	Headphones	110	3

Insight 3: The highest-spending customers are disproportionately outside the loyalty program, though this may reflect missing data rather than a true behavioral pattern.

Of the top 11 customers by total spend, only one (customer 12, ranked #2) is a confirmed loyalty member. However, 6 of the 11 show "None" rather than an explicit "No," which lines up with the missing and placeholder values in the loyalty program field identified earlier in the dataset. Before treating this as a real finding, such as "top spenders are not loyalty members," it is worth confirming whether "None" reflects confirmed non-membership or unknown data, since those support two very different business conclusions. Customer 13 is the single highest spender overall at \$440 across 8 purchases.

Customer ID	Loyalty	Purchases	Total Spent	Avg Order	Rank
13	None	8	\$440.00	\$55.00	1
12	Yes	6	\$400.00	\$66.67	2
55	None	6	\$350.00	\$58.33	3
51	None	5	\$300.00	\$60.00	4

SQL Queries

Insight 1: Employee Performance by Location

```
WITH emp_sales AS (  
    SELECT store_location,  
           AVG(CAST(sales_performance AS FLOAT)) as avg_sales,  
           COUNT(*) as employee_count  
    FROM Employee_Records  
    WHERE sales_performance IS NOT NULL  
    GROUP BY store_location  
)  
SELECT store_location, avg_sales, employee_count,  
       RANK() OVER (ORDER BY avg_sales DESC) as location_rank  
FROM emp_sales  
ORDER BY avg_sales DESC
```

Insight 2: Top-Selling Products by Category

```
SELECT *  
FROM (  
    SELECT p.category,  
           p.product_name,  
           SUM(s.quantity) AS total_sold,  
           RANK() OVER (  
               PARTITION BY p.category  
               ORDER BY SUM(s.quantity) DESC  
           ) AS rank_in_category  
    FROM Sales_Transactions s  
    JOIN Product_Details p ON s.product_id = p.product_id  
    GROUP BY p.category, p.product_name  
)  
WHERE rank_in_category <= 3  
ORDER BY category, total_sold DESC
```

Insight 3: Customer Purchasing Behavior

```
SELECT
    c.customer_id,
    c.loyalty_program,
    COUNT(s.transaction_id) AS purchase_count,
    ROUND(SUM(s.total_amount), 2) AS total_spent,
    ROUND(AVG(s.total_amount), 2) AS average_order_value,
    RANK() OVER (
        ORDER BY SUM(s.total_amount) DESC
    ) AS spending_rank
FROM Customer_Demographics c
JOIN Sales_Transactions s
    ON c.customer_id = s.customer_id
WHERE s.total_amount IS NOT NULL
GROUP BY
    c.customer_id,
    c.loyalty_program
ORDER BY total_spent DESC;
```

Data Cleaning and Process Documentation

Employee Records

Before analyzing sales performance, I ran a diagnostic query to quantify how much of the sales_performance field was missing or stored as the text placeholder "nan":

```
SELECT COUNT(*) as total_rows,
    SUM(CASE WHEN sales_performance IS NULL OR sales_performance = 'nan' THEN 1 ELSE 0 END) as
missing_sales
FROM Employee_Records
```

This confirmed the scale of the missing-data problem before I applied the CAST and IS NOT NULL filtering used in the Insight 1 query, rather than assuming the cleaning approach without first measuring it.

Product Details

I used a conservative cleaning approach rather than dropping entire rows or inventing values. Non-numeric text such as "one thousand" was converted to its numeric equivalent, 1000. "Nan" price entries were replaced with SQL NULL. Missing price and stock values were left as NULL instead of imputed, since no reliable business rule was provided for estimating them. Those missing values were excluded only from calculations that specifically required price or stock data, which preserved the underlying product records while preventing invalid values from distorting the analysis.

Customer Demographics

Missing age and loyalty program values were retained rather than imputed, since there was insufficient information to determine the correct values. Text placeholders such as "nan" and "N/A" were standardized to SQL NULL, and records with those NULLs were excluded only from analyses that specifically required those fields. This decision directly explains the "None" values seen in the Insight 3 output, and is also why that insight is presented with a caveat about data quality rather than as a confirmed behavioral pattern.

Sales Transactions

Missing transaction values were excluded where required, but the initial pass through this table did not fully standardize text-based numeric entries. A subsequent cleaning pass converted known number words such as "three" to their numeric equivalents, standardized "nan" and "N/A" to NULL, and cast valid quantities and amounts to the appropriate numeric types before they were used in aggregation.

Challenges

The hardest part of the process was cleaning numeric fields that contained a mixture of valid numbers, text-based numbers, and missing-value placeholders in the same column. Values such as "five thousand," "one thousand," and "three" could not be analyzed reliably until they were converted using CASE statements, while "nan" and "N/A" needed to become SQL NULL. Getting the CASE logic right took several attempts to make sure valid values were preserved and invalid entries were handled correctly rather than silently dropped or miscalc. I also learned that data-changing queries such as UPDATE must be executed directly through the database connection rather than through `pd.read_sql_query()`, which is built to return query results as a DataFrame rather than to execute write operations.

Areas for Further Analysis

- Investigate whether Chicago's high average reflects fewer, stronger performers or a smaller-denominator effect, and whether Houston's staffing level relative to output signals a training or scheduling gap.
- Layer in `sale_date` to see whether product rankings hold steady across seasons or shift, which is directly relevant to TechMart's inventory planning.
- Resolve the loyalty program ambiguity in `Customer_Demographics`, then rerun the customer spending analysis to determine whether top spenders are truly opting out of the program or are simply missing enrollment data.
- Cross-reference top-selling products against store location to test whether Chicago's high performance is product-mix driven rather than staffing driven.

Conclusion

Across all three analyses, the same theme recurs: raw rankings tell only part of the story until they are checked against data quality and context. Chicago's top ranking in employee performance invites a staffing and product-mix investigation rather than a simple conclusion. Product rankings show a meaningful difference in demand concentration between categories. Customer spending patterns are promising but currently confounded by inconsistent loyalty program data. Each of these findings points toward a clear, specific next step rather than a dead end, which is the intended outcome of this analysis.